

Software Engineering

Exercise and Lab: Advanced Class Diagram

1. Class diagram - Vehicle

All Vehicles have some common attributes (e.g speed and colour) and common behaviour (turnLeft, turnRight).

Bicycle and MotorVehicle are both kinds of Vehicle and are therefore considered as subclasses of the class Vehicle.

In the class MotorVehicle, we have some attributes to indicate the number of engines and license plates, and it allows us to examine these attributes accordingly. For the class Bicycle, there is only one operation, which is used to ring the bell.

MotorVehicles is the base class of another two classes called MotorBike and Car. For the class MotorBike, we could perform the method revEngine() while the class Car contains the number of doors and allows us to switch on the air conditioner and get the number of doors in a car.

Draw a class diagram to represent the relationships of the Vehicle by considering the information mentioned above.

2. Class diagram - Car

One car is characterized by its marque, model, and motorization. We supposed that the marque of car could be from the elements {Citroen, Fiat, Ford, Nissan, Toyota, Renault}. For the motorization, we can categorize by the engine of the car containing its power and designation.

One car contains 4 wheels and the engine operates two of them.

Draw a class diagram to represent this system.

3. Simple School Management System

A small school would like to build a system to manage over learning operations such as:

- a. Person represents all stakeholders including Teachers and Students.
Each Person has:
 - Address (a class containing Street, City, State, Postal Code, and Country), we can check validity of an address
 - A name, phone number and email
 - Everyone can buy parking card (a card used for monthly payment for parking service)
- b. Student represents all students in this school. Student is a Person.
Each student contains
 - Student ID

- Average score
 - Group (eg. A class containing group name (A,B...), and an ID)
- c. Teacher represents all teachers in this school. Teacher is a Person and contains skills.

Draw Class diagram of the system.